**Oluwatosin Ayeni**

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**Perplexity AI**

Dear Hiring Team,

I am applying for the Member of Technical Staff (AI Inference Engineer) role at Perplexity AI. Perplexity's product is only as good as the inference latency behind it — every query is a race between model quality and response time, and I have spent the last two years making that race faster.

Three achievements that speak directly to Perplexity's inference requirements:

- **H100 NVL — latency at the microsecond level:** I benchmarked vLLM v0.30 vs SGLang v0.5.20 on 2× NVIDIA H100 NVL 94 GB (NVLink 4.0, CUDA 13.4), Qwen3-8B BF16, TP=2. Peak throughput: 5,965 tok/s. Critical finding: RadixAttention prefix caching reduced TTFT from 55,565 ms to 39 ms — a 1,443× speedup — at 131K-token context with 100% SLO compliance at concurrency 128. Full benchmark: inference-web-theta.vercel.app/benchmarks/llm.
- **Production inference gateway — 13 optimisations:** I built *inference-optimizer*, a production-grade AI Inference Gateway across Claude, GPT-4o, and Gemini: parallel execution (3.87× faster), prompt caching (61.4% cost reduction), model routing (64.3% cheaper), dynamic concurrency (14× throughput). The same architecture patterns apply directly to high-throughput query serving at Perplexity's scale.
- **GPU profiling and CUDA kernel work:** Using NVIDIA Nsight Systems and Compute, I diagnosed a memory bandwidth bottleneck in KV cache CUDA ops and retuned the kernel, delivering 3.8× throughput improvement. Also achieved 18× latency reduction through continuous batching pipeline redesign.

Perplexity's engineering challenge is unique: sub-second answers over a live web index using frontier models, at millions of queries per day. That is the class of inference problem I find most interesting — where the gap between a fast system and a slow one is the product itself. I look forward to hearing from you.

Yours sincerely,

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